

Properties of Limits (1.5)

Today's Target

"I can apply any limit property, when it is appropriate. To evaluate a limit, I always apply direct substitution first. However, if this results in an indeterminate result, I apply an appropriate technique to determine the limiting value, if it exists. I can apply what I have learned in familiar and unfamiliar settings."



A1. Rates of Change



Limits (3-cases)

$$\lim_{x \rightarrow a} f(x)$$

Peicewise

$$f(x) = \begin{cases} \sim & x > a \\ \sim & x = a \\ \sim & x < a \end{cases}$$

$$\lim_{x \rightarrow a^-} f(x)$$

$$\lim_{x \rightarrow a^+} f(x)$$

if $=$ then limit exists

Sub $x=a$ into $f(x)$

$f(a)$ is undefined

use limit solving methods

Sub it in to the simplified f^D

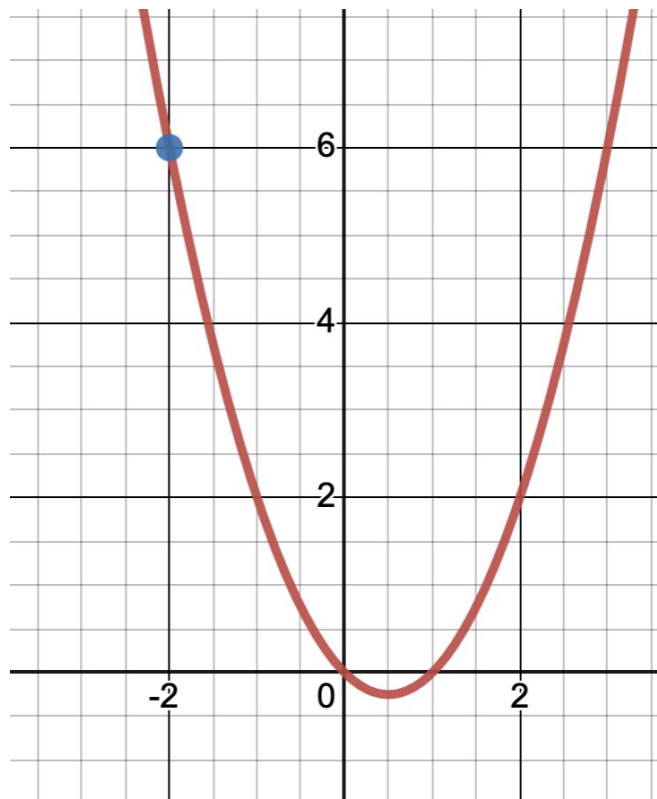
$f(a)$ is defined

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Haadiyah
Ethan
Kilmer

Example 1

Evaluate $\lim_{x \rightarrow -2} (x^2 - x)$. Verify graphically in [desmos](https://www.desmos.com/calculator)



Properties of Limits

For any real number a , suppose that f and g both have limits that exist at $x = a$.

1. $\lim_{x \rightarrow a} k = k$, for any constant k
2. $\lim_{x \rightarrow a} x = a$
3. $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$
4. $\lim_{x \rightarrow a} [cf(x)] = c[\lim_{x \rightarrow a} f(x)]$, for any constant c
5. $\lim_{x \rightarrow a} [f(x)g(x)] = [\lim_{x \rightarrow a} f(x)][\lim_{x \rightarrow a} g(x)]$
6. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$, provided that $\lim_{x \rightarrow a} g(x) \neq 0$
7. $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$, for any rational number n

Evaluate the limit using the properties of limits

$$\lim_{x \rightarrow 2} \frac{f(x) + x^2}{[g(x)]^2}$$

Given

$$\lim_{x \rightarrow 2} f(x) = 15$$

$$\lim_{x \rightarrow 2} g(x) = -1$$

in the library - and

the limits of f and g exist @ $x=2$

$$\lim_{x \rightarrow 2} \frac{f(x) + x^2}{[g(x)]^2}$$

Quotient (6)

$$= \frac{\lim_{x \rightarrow 2} (f(x) + x^2)}{\lim_{x \rightarrow 2} [g(x)]^2}$$

Sum

$$= \frac{\lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} x^2}{\left[\lim_{x \rightarrow 2} g(x) \right]^2}$$

Power

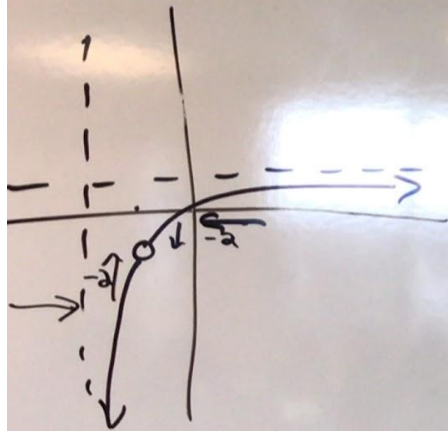
$$= \frac{15 + \left[\lim_{x \rightarrow 2} x \right]^2}{(-1)^2}$$

power

$$= 15 + 2^2$$

linear

$$= 19$$



$$\begin{aligned}
 & \lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2} \\
 &= \lim_{x \rightarrow -1} \frac{(x-1)\cancel{(x+1)}}{\cancel{(x+1)}(x+2)} \\
 &= \lim_{x \rightarrow -1} \frac{x-1}{x+2} \\
 &= -2
 \end{aligned}$$

$\swarrow$ cancel $x \neq -1, -2$
 $\swarrow$ sub in $x = -1$

$$\begin{aligned}
 f(x) &= \frac{x-1}{x+2} \quad x \neq -1, -2 \\
 &\quad \text{point disc. } (-1, -2) \\
 f(-1) &= \frac{-2}{1} \\
 &= -2
 \end{aligned}$$

Let

$$\begin{aligned}
 f(x) &= \frac{x^2 - 1}{x^2 + 3x + 2} \\
 f(-1) &= \frac{0}{0} \\
 &\quad \downarrow \\
 &\text{undefined}
 \end{aligned}$$

$$f(x) = x$$

$$\begin{aligned} \lim_{x \rightarrow 4} x^{\frac{1}{2}} \\ &= \left(\lim_{x \rightarrow 4} x \right)^{\frac{1}{2}} \\ &= 4^{\frac{1}{2}} \\ &= \sqrt{4} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -2} x^{\frac{1}{2}} \\ &= \left(\lim_{x \rightarrow -2} x \right)^{\frac{1}{2}} \\ &= (-2)^{\frac{1}{2}} \\ &= \text{The limit} \\ &\quad \text{does not} \\ &\quad \text{exist} \end{aligned}$$

Wipebook

$$\begin{aligned} f(x) &= x^2 \\ \lim_{x \rightarrow -2} [f(x)]^{\frac{1}{2}} \\ &= \left[\lim_{x \rightarrow -2} x^2 \right]^{\frac{1}{2}} \\ &= [(-2)^2]^{\frac{1}{2}} \\ &= 4^{\frac{1}{2}} \\ &= 2 \end{aligned}$$

$$\lim_{x \rightarrow -2} (x^2 - x)$$

$$= \lim_{x \rightarrow -2} x(x-1)$$

$$= \left[\lim_{x \rightarrow -2} x \right] \left[\lim_{x \rightarrow -2} (x-1) \right] \quad \# 5$$

$$= \left[\lim_{x \rightarrow -2} x \right] \left[\lim_{x \rightarrow -2} x - \lim_{x \rightarrow -2} 1 \right] \quad \# 3$$

$$= [-2] [-2 - 1] \quad \# 1$$

$$= (-2)(-3) = 6$$

Indeterminate vs. Undefined

$\frac{0}{0}$ is considered an example of an indeterminate form, whereas $\frac{1}{0}$ is considered an example of an undefined form. Another example of an undefined result is $\sqrt{-1}$

Recall: Absolute value (MHF 4UI)

$$|a| = \left\{ \begin{array}{ll} a & a \geq 0 \\ -a & a < 0 \end{array} \right\}$$

At the boards- with your groups

Example 3

Evaluate, if it exists:

a) $\lim_{x \rightarrow 3} \frac{9 - x^2}{3 - x}$

Factoring Strategy

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{9 - x^2}{3 - x} \\ \lim_{x \rightarrow 3} \frac{(3+x)(3-x)}{(3-x)} \quad x \neq 3 \\ \lim_{x \rightarrow 3} 3+x \\ = 6 \end{aligned}$$

$$\begin{aligned} \text{Let } f(x) &= 9 - x^2 \\ \text{Let } g(x) &= 3 - x \\ \lim_{x \rightarrow 3} \frac{9 - x^2}{3 - x} \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &\rightarrow \frac{\lim_{x \rightarrow a} 9 - x^2}{\lim_{x \rightarrow a} 3 - x} \\ &\Rightarrow \lim_{x \rightarrow a} 3 - x \neq 0 \\ &\quad \lim_{x \rightarrow a} \neq 3 \\ &= \frac{\lim_{x \rightarrow a} (3+x)(3-x)}{\lim_{x \rightarrow a} 3-x} \\ &= \lim_{x \rightarrow 3} 3+x \\ &= 6 \end{aligned}$$

b) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 + 3x}}{x}$

Rationalize strategy

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1+3x}}{x} \cdot \frac{1 + \sqrt{1+3x}}{1 + \sqrt{1+3x}}$$

$$\lim_{x \rightarrow 0} \frac{1 - 1 - 3x}{x(1 + \sqrt{1+3x})}$$

$$\lim_{x \rightarrow 0} \frac{-3x}{x(1 + \sqrt{1+3x})}$$

$$\lim_{x \rightarrow 0} \frac{-3}{1 + \sqrt{1+3x}}$$

$$= \frac{-3}{2}$$

c) $\lim_{x \rightarrow 2} \frac{x |x - 2|}{x - 2}$

Definition of absolute value strategy

$$c) \quad \lim_{x \rightarrow 2} \frac{x|x-2|}{x-2}$$

$$\begin{array}{l} x|x-2| \\ \swarrow \quad \searrow \\ x \geq 2 \quad x < 2 \\ x^2 - 2x \quad -x^2 + 2x \end{array}$$

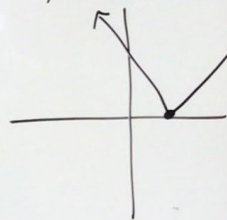
Case #1 $x \geq 2$

$$\begin{aligned} \lim_{x \rightarrow 2^+} \frac{x(\cancel{x-2})}{(\cancel{x-2})} \\ = \lim_{x \rightarrow 2^+} x \\ = 2 \end{aligned}$$

Case $x < 2$

$$\begin{aligned} \lim_{x \rightarrow 2^-} \frac{x(-)(\cancel{x-2})}{(\cancel{x-2})} \\ = \lim_{x \rightarrow 2^-} \frac{-x(\cancel{x-2})}{(\cancel{x-2})} \\ = \lim_{x \rightarrow 2^-} -x \\ = -2 \\ = -2 \end{aligned}$$

Consider $|x-2|$



The limit does not exist

$$|x-2| = \begin{cases} x-2 & x \geq 2 \\ -(x-2) & x < 2 \end{cases}$$

d)

$$\lim_{x \rightarrow 0} \frac{(x+8)^{\frac{1}{3}} - 2}{x}$$

Substitution method

$$\lim_{x \rightarrow 0} \frac{(x+8)^{\frac{1}{3}} - 2}{x}$$

$$= \lim_{u \rightarrow 2} \frac{u-2}{u^3-8}$$

$$= \lim_{u \rightarrow 2} \frac{\cancel{u-2}}{(\cancel{u-2})(u^2+2u+4)}$$

$$= \lim_{u \rightarrow 2} \frac{1}{u^2+2u+4}$$

$$\text{Let } u = (x+8)^{\frac{1}{3}}$$

$$u^3 = x+8$$

$$u^3 - 8 = x$$

$$\text{If } u(x) = (x+8)^{\frac{1}{3}}$$

$$\lim_{x \rightarrow 0} u(x) = (0+8)^{\frac{1}{3}} = 2$$

$$\text{As } x \rightarrow 0 \\ u \rightarrow 2$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$u^2 + 2u + 4$$

$$= \frac{1}{12}$$

You will "absolutely" love these questions!

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Level

1	4e	8def	9ade	10*ad	13**	15	16
2	1	4	1,4,4	3	4	4+	4+

*Sketches are not required.

**Have you understood Example 2 from the lesson?

Review it on your own: [CEMC](#)

d) $\lim_{x \rightarrow 1} \frac{x^{\frac{1}{6}} - 1}{x - 1}$